

Lecture 18

Sequential Circuits and Latches

Sequential circuits · synchronous and asynchronous · the SR and S'R' latches · gated latches · the D latch and its timing

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 5, §5.1–5.3



State: the circuit remembers

Today's roadmap

1

§5.1–5.2

Sequential

Memory and the clock

2

§5.3

SR latches

Two cross-coupled gates

3

§5.3

Gated latches

SR and D with an enable

4

§5.3

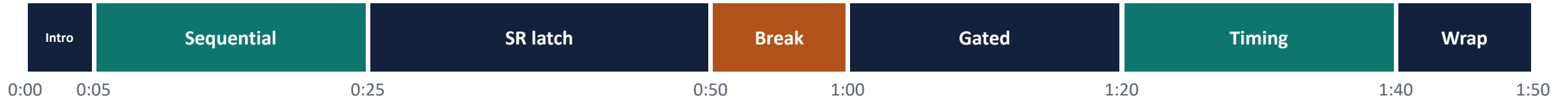
Timing

Transparency and its risk

By the end of class you can...

- Explain what makes a circuit sequential
- Distinguish synchronous from asynchronous behavior
- Trace SR (NOR) and S'R' (NAND) latches
- Build a D latch and draw its timing diagram
- Explain why a transparent latch is risky in a clocked system

Agenda for today's session

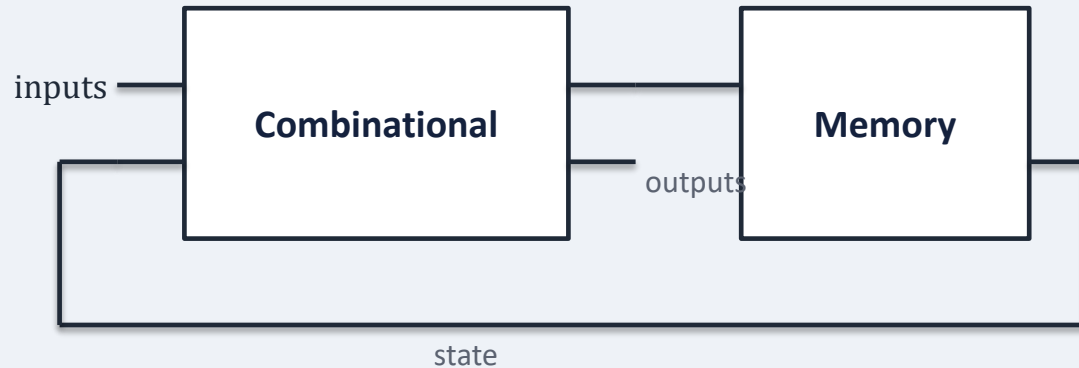


Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:25	Sequential circuits and the clock	—
0:25 – 0:50	SR and S'R' latches	Example 1 · Your turn 1
0:50 – 1:00	Break	
1:00 – 1:20	Gated SR latch; the D latch	—
1:20 – 1:40	Latch timing and transparency	Example 2 · Your turn 2
1:40 – 1:50	Symbols; concept check; wrap-up	Lecture 19 preview

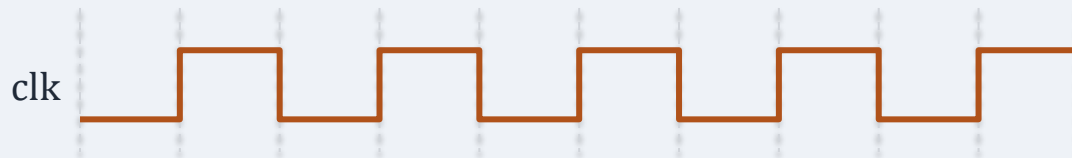
Sequential circuits

§5.1–5.2

Block diagram



The clock



State may change only at clock edges in a synchronous circuit.

Sequential

Outputs depend on the inputs and on the stored state.
Described by a state table or state diagram, not a truth table alone.

Synchronous

State changes only at discrete instants set by a clock.

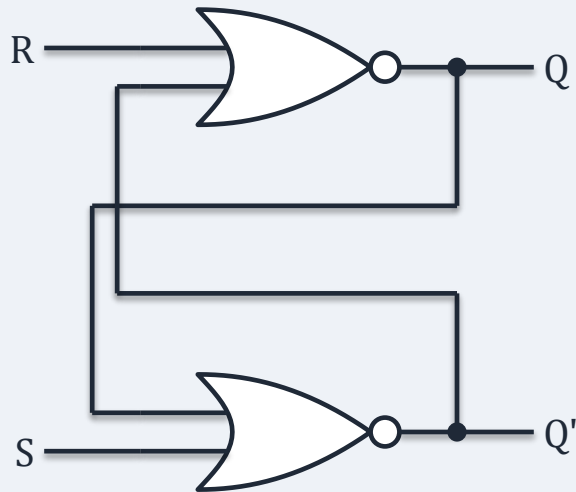
Asynchronous

State changes whenever inputs change: fast but hard to design safely.

The SR latch (NOR)

§5.3

Two cross-coupled NOR gates



S	R	Q	Q'	Action
1	0	1	0	set
0	0	1	0	hold (after set)
0	1	0	1	reset
0	0	0	1	hold (after reset)
1	1	0	0	forbidden

Two stable states

With $S = R = 0$ the latch holds whichever state it was in: one bit of memory. $S = R = 1$ forces $Q = Q' = 0$ and is not allowed.

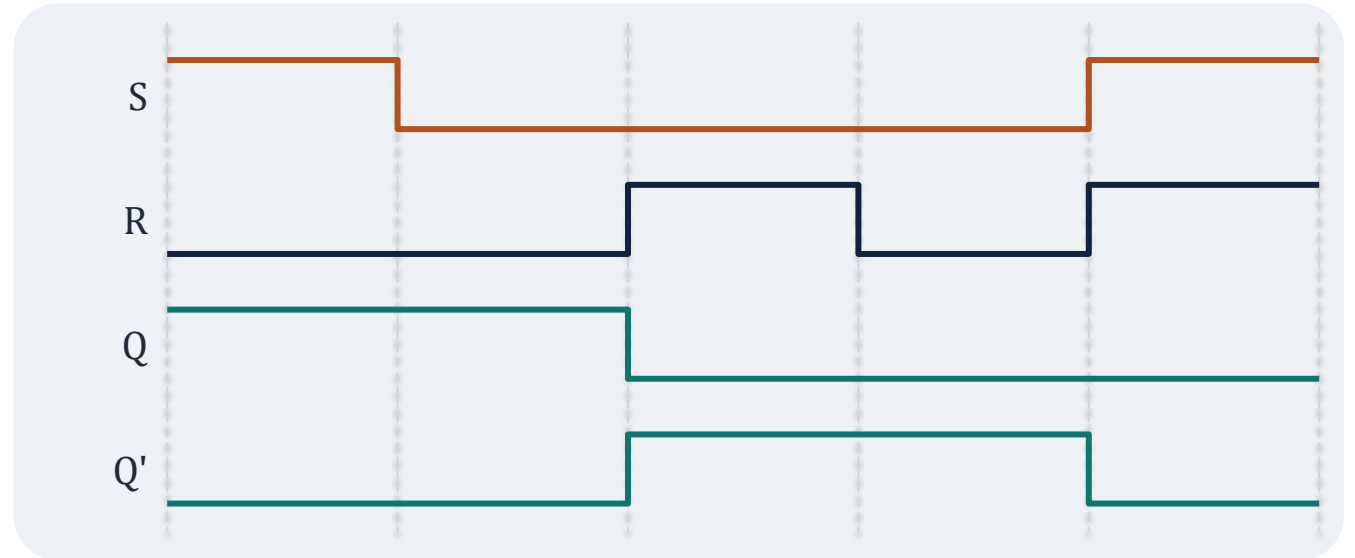
Example 1 — Trace an SR latch

§5.3

EXAMPLE

An SR (NOR) latch starts with $Q = 0$. Apply $SR = 10, 00, 01, 00, 11$ in turn.

Step	S	R	Q	Q'	Action
1	1	0	1	0	set
2	0	0	1	0	hold
3	0	1	0	1	reset
4	0	0	0	1	hold
5	1	1	0	0	forbidden

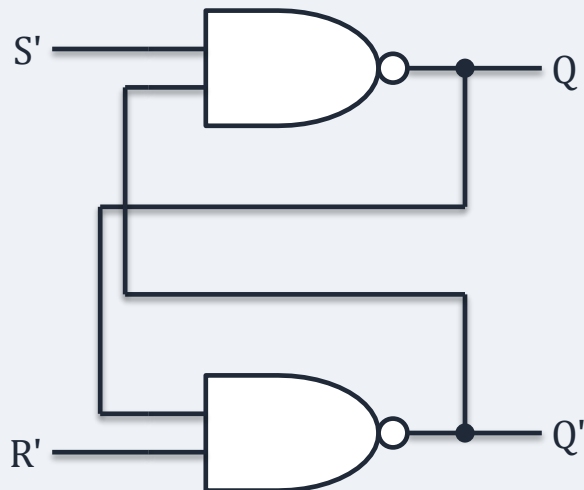


- 1 $S = 1$ sets, $R = 1$ resets, 00 holds
- 2 Step 1: $10 \rightarrow$ set: $Q = 1, Q' = 0$
- 3 Step 2: $00 \rightarrow$ hold: Q stays 1
- 4 Step 3: $01 \rightarrow$ reset: $Q = 0, Q' = 1$
- 5 Step 4: $00 \rightarrow$ hold: Q stays 0
- 6 Step 5: $11 \rightarrow Q = Q' = 0$: forbidden

The $S'R'$ latch (NAND)

§5.3

Two cross-coupled NAND gates



S'	R'	Q	Q'	Action
0	1	1	0	set
1	1	1	0	hold (after set)
1	0	0	1	reset
1	1	0	1	hold (after reset)
0	0	1	1	forbidden

Active-low inputs

A 0 on S' sets, a 0 on R' resets, 11 holds. It is the NOR latch with every input inverted.

Your turn 1 — Trace a NAND latch

§5.3

YOUR TURN

≈ 8 min

An $S'R'$ (NAND) latch starts with $Q = 0$. Apply $S'R' = 11, 01, 11, 10, 11, 00$ in turn.

(a) Give Q and Q' after each step and name the action.

(b) Which step is not allowed, and why?

(c) Which steps show the latch remembering?

Remember: the inputs are active low.

Your turn 1 — Solution

§5.3

SOLUTION

Step	S'	R'	Q	Q'	Action
1	1	1	0	1	hold
2	0	1	1	0	set
3	1	1	1	0	hold
4	1	0	0	1	reset
5	1	1	0	1	hold
6	0	0	1	1	forbidden

- 1 NAND latch: active low. $S' = 0$ sets, $R' = 0$ resets, 11 holds
- 2 Step 1: 11 $\rightarrow$ hold: Q stays 0
- 3 Step 2: 01 $\rightarrow S' = 0$ sets: $Q = 1$
- 4 Step 3: 11 $\rightarrow$ hold: $Q = 1$
- 5 Step 4: 10 $\rightarrow R' = 0$ resets: $Q = 0$
- 6 Step 5: 11 $\rightarrow$ hold: $Q = 0$
- 7 Step 6: 00 $\rightarrow$ both NANDs give 1: $Q = Q' = 1$, **forbidden**
- 8 (b) Releasing both to 1 races: the next state is unknown
- 9 (c) Steps 1, 3 and 5 hold the stored bit

Break

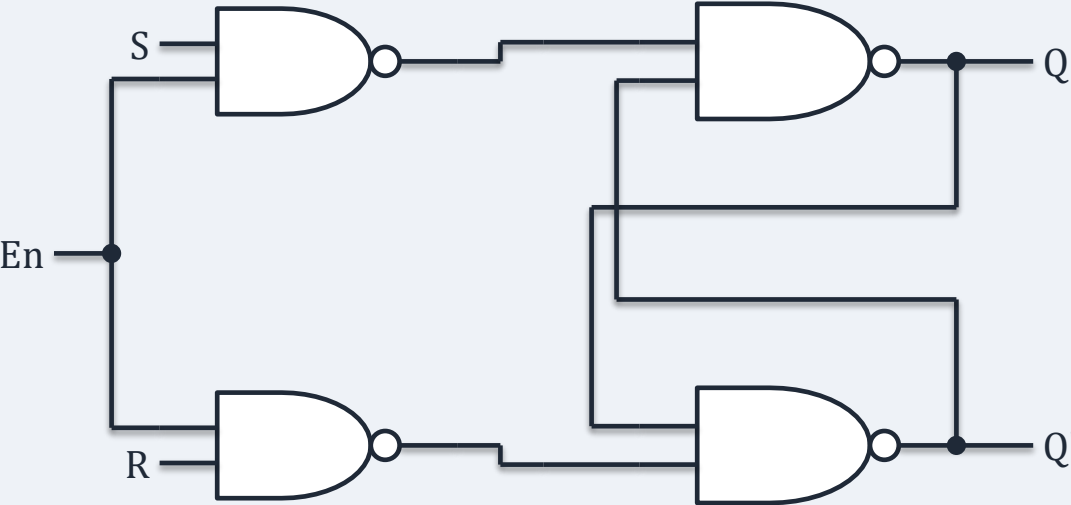
10 minutes — we resume at the 1:00 mark

Next: gated latches and the D latch

SR latch with a control input

§5.3

NAND gates steer S and R into an S'R' latch



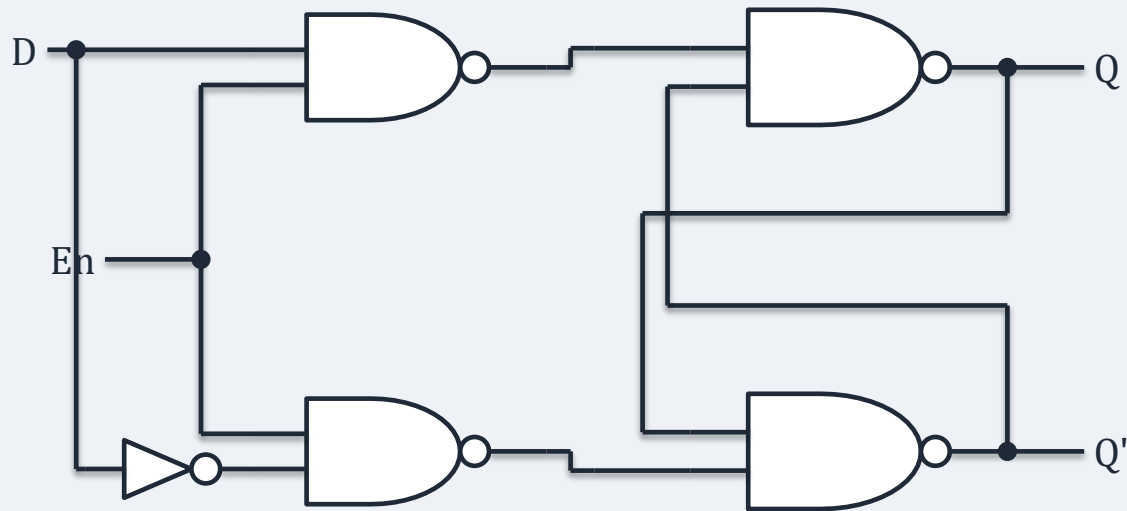
En	S	R	Next state
0	X	X	no change
1	0	0	no change
1	0	1	Q = 0: reset
1	1	0	Q = 1: set
1	1	1	forbidden

With $En = 0$ both steering gates output 1: the latch holds no matter what S and R do. $En = 1$ makes the inputs active high.

The D latch

§5.3

One data input: no forbidden state



En	D	Next state
0	X	no change
1	0	$Q = 0$
1	1	$Q = 1$

Transparent

While $En = 1$, Q follows D : the latch is a wire.

When En falls, Q keeps the last value of D .

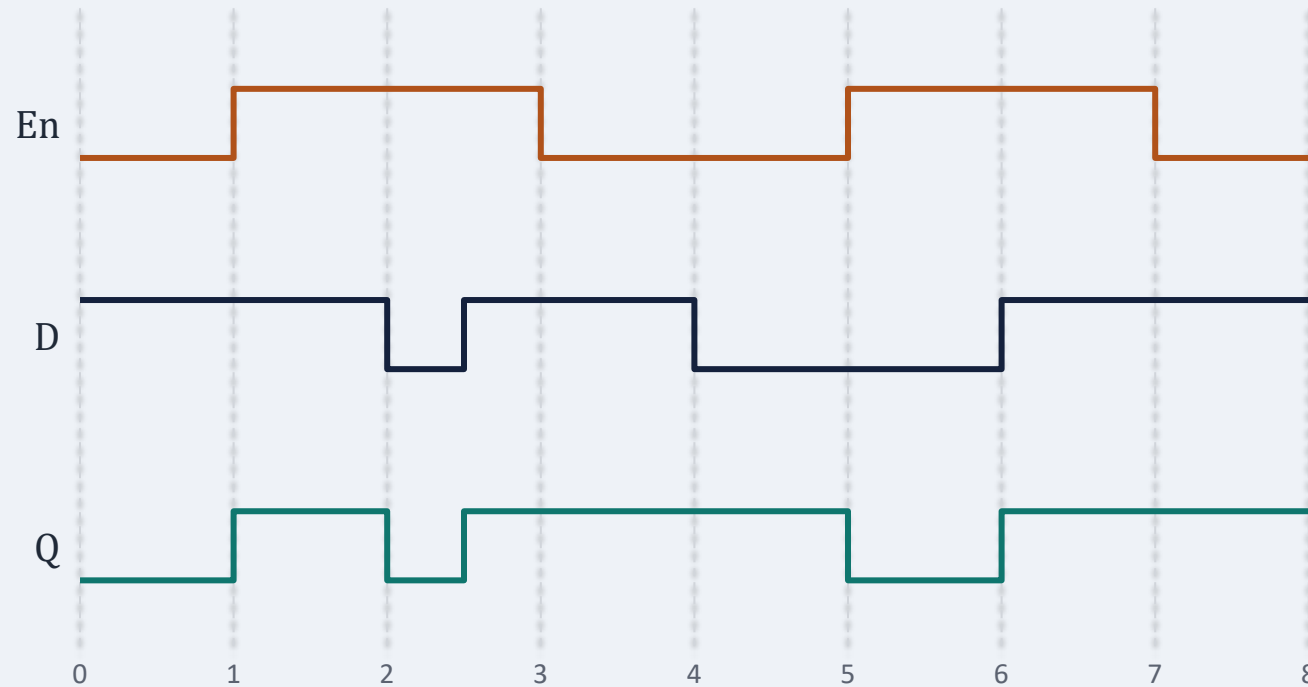
S and R can never both be 1: the inverter sees to it.

Example 2 — D latch timing

§5.3

EXAMPLE

Given En and D , draw Q for a D latch that starts with $Q = 0$.



- 1 Rule: $En = 1 \rightarrow$ follow D ; $En = 0 \rightarrow$ hold
- 2 0–1: $En = 0$: Q holds 0
- 3 At 1: En rises with $D = 1 \rightarrow Q = 1$
- 4 2–2.5: D dips $\rightarrow Q$ follows: 0, then 1
- 5 At 3: En falls $\rightarrow Q$ holds 1
- 6 At 4: D falls, but $En = 0 \rightarrow$ ignored
- 7 At 5: En rises with $D = 0 \rightarrow Q = 0$
- 8 At 6: $Q = 1$; at 7 En falls: holds 1

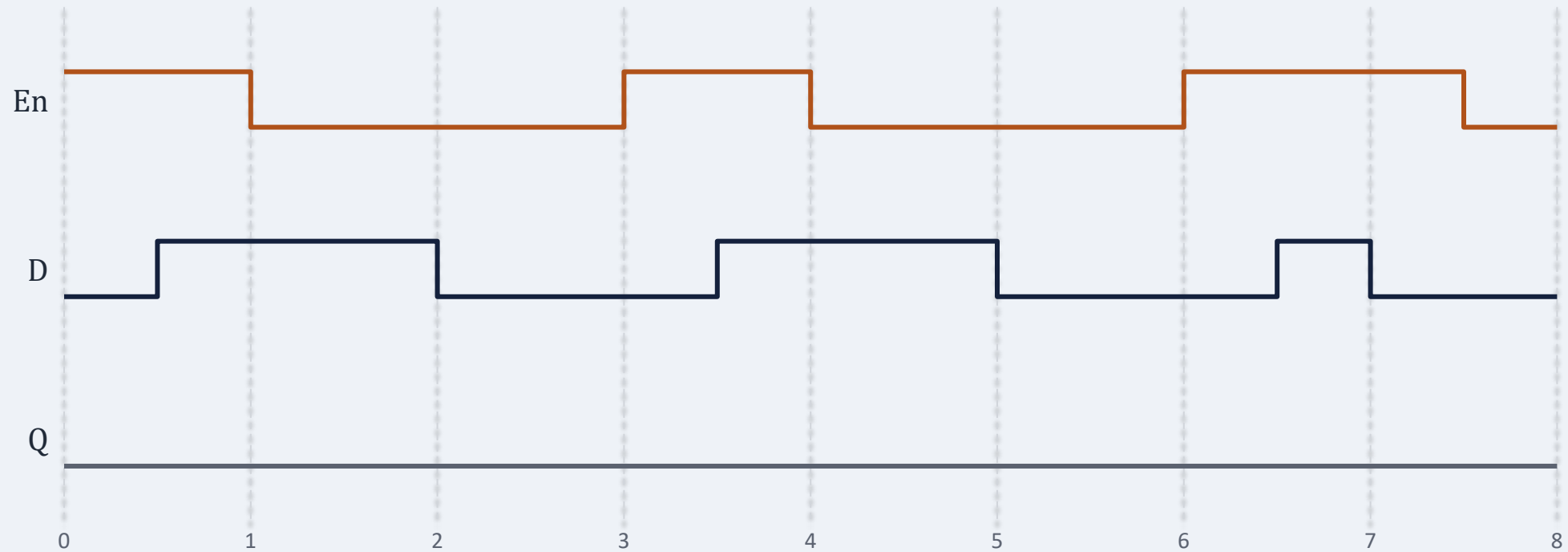
Your turn 2 — Draw Q

§5.3

YOUR TURN

≈ 8 min

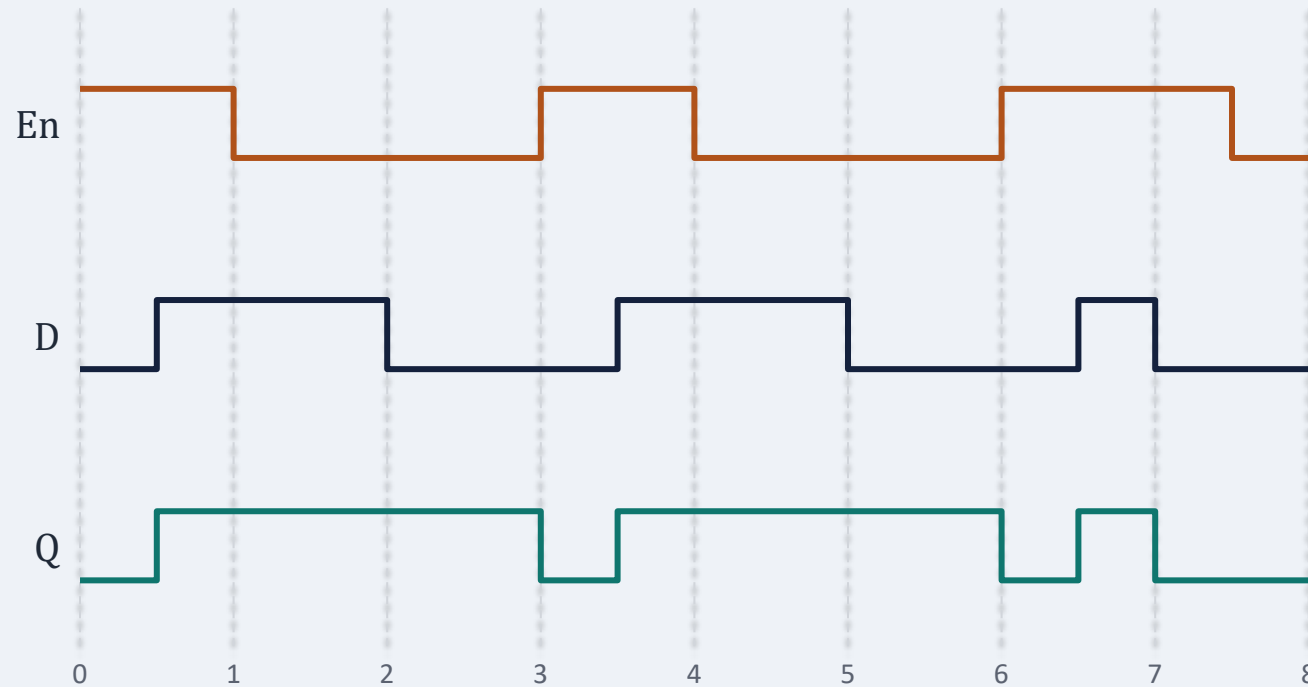
A D latch starts with $Q = 0$. Draw Q .



Your turn 2 — Solution

§5.3

SOLUTION

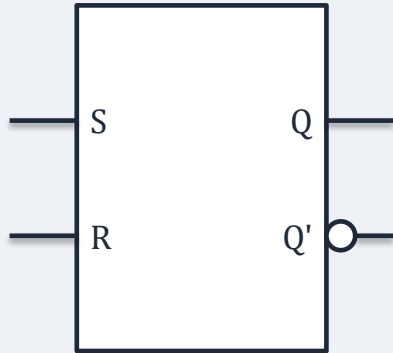


- 1 Rule: $En = 1 \rightarrow$ follow D ; $En = 0 \rightarrow$ hold
- 2 0–1: $En = 1$: $Q = D$: 0, then 1 at 0.5
- 3 At 1: En falls $\rightarrow Q$ holds 1
- 4 At 2: D falls, $En = 0 \rightarrow$ ignored
- 5 At 3: En rises with $D = 0 \rightarrow Q = 0$
- 6 At 3.5: $Q = 1$; at 4 En falls: holds 1
- 7 At 5: D falls, $En = 0 \rightarrow$ ignored
- 8 6–7.5: Q follows D : 0, 1 at 6.5, 0 at 7
- 9 At 7.5: En falls $\rightarrow Q$ holds 0

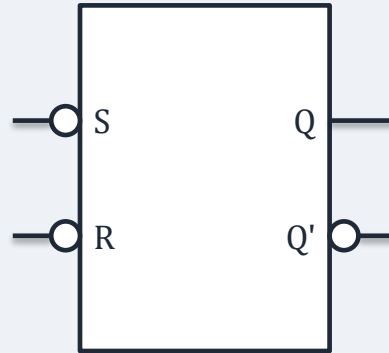
Latch symbols

§5.3

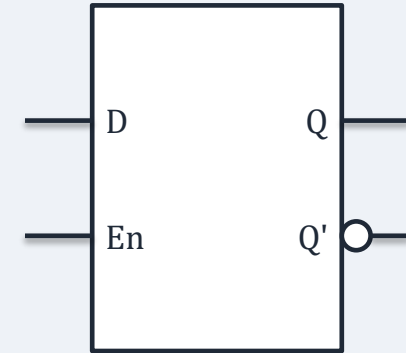
SR latch



S'R' latch



D latch



Why latches are not enough

If a latch output feeds back through logic to its own input while enabled, it can change again and again within one pulse. Flip-flops respond only at a clock edge (Lecture 19).

Concept check

1

What makes a circuit sequential rather than combinational?

2

What does the SR latch do when $S = R = 0$?

3

Why is $S = R = 1$ forbidden in the NOR latch?

4

Why can a D latch never enter the forbidden state?

Key takeaways

Sequential

$\text{outputs} = f(\text{inputs}, \text{state})$

Synchronous

change at clock edges

SR (NOR)

$S = R = 1$ forbidden

S'R' (NAND)

active-low inputs

D latch

$Q = D$ while $En = 1$

Transparent

the risk flip-flops fix

Two cross-coupled gates store one bit.

Homework #8 due at the start of Lecture 19

This afternoon's lab: Lab Hp2 — Calculator 2: 2-bit adder-subtractor

Next: flip-flops and sequential circuit analysis